Fiscal Year:	FY 2014	**Task Last Updated:**	FY 01/25/2022
PI Name:	Krebs, Christopher M.S.		
Project Title:	Enhanced Dynamic Load Sensor for ISS (EDLS-ISS)		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Cambridge	**State:**	MA
Zip Code:	02142-1401	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	SBIR Phase II
Start Date:	05/30/2014	**End Date:**	05/29/2016
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX14CS55C		
Performance Goal No.:			
Performance Goal Text:			

Aurora Flight Sciences and the Massachusetts Institute of Technology (MIT) propose to develop a stand-alone 6-degree of freedom (DOF) load sensing system that collects biomechanical force and moment data while integrated with the Advanced Resistive Exercise Device (ARED) on the International Space Station (ISS) by adapting and enhancing previously field-tested systems. Our Enhanced Dynamic Load Sensors for ISS (EDLS-ISS) system includes a separate base unit tethered to two low-profile sensors, which can be non-invasively mounted on the surface of the ARED to serve as new footplates.
The EDLS-ISS is being developed for two reasons:

- Operational. Force plates on the ARED will allow crewmembers to verify loading and to record forces. This device provides a simple, non-invasive solution that measures foot forces and displays this to the astronaut.

- Research. Kinematic experiments on the ARED, for example, characterizing limb segment relative motions during a

Task Description:	squat, require corresponding force and moment measurements. Our EDLS-ISS system provides three-axis force and three-axis moment measurements for such experiments. By providing a short-term operational solution and a sustainable long-term research solution, ELDS-ISS is an important tool for the spaceflight community. During Phase I, we improved the design of the existing 6-DOF Microgravity Investigation of Crew Reactions in 0-G (MICR0-G) Adapt sensor module, previously developed by Aurora and MIT. These improvements optimized the sensor for use on the ARED: decreasing the overall height of the sensors and separating the electronics into an off-board "base unit," increasing the footprint size to cover most of the surface of the ARED platform, updating the hard drive and electronics, providing a real-time user-feedback display, and developing ARED mounting options. The anticipated result of our Phase II effort will be three complete EDLS-ISS systems, each of which includes a base unit and two sensor units. Two systems will be flight-qualified.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	New project for FY2014. Reporting not required for this SBIR Phase II project. (Ed. note: added to Task Book on 01/20/2022)
Bibliography Type:	Description: (Last Updated:)